

1 Skeletal muscle is made of bundles of fibres.

(a) Describe the role of calcium ions in producing contraction of a muscle fibre.

(2)

(b) The table shows some features of slow and fast twitch muscle fibres.

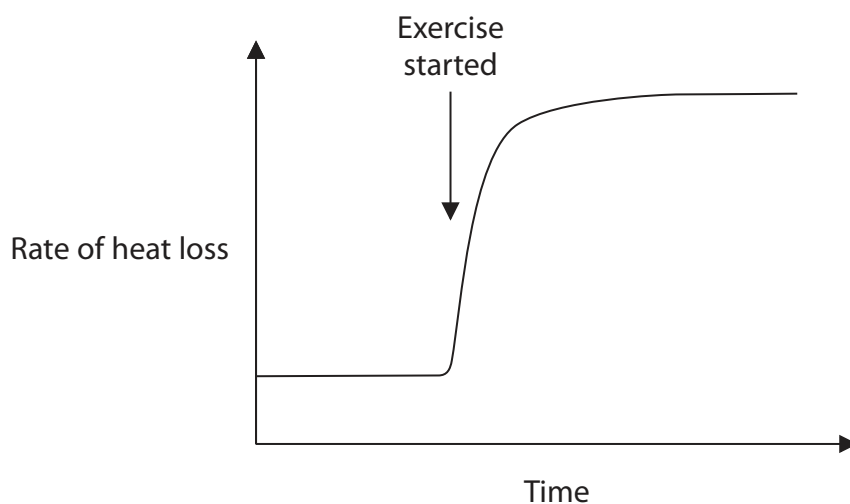
Feature of fibre	Slow twitch fibre	Fast twitch fibre
Number of mitochondria per fibre	Many	Few
Number of myoglobin molecules per fibre	Many	Few
Capillary network	Larger	Smaller
Contraction	Slow and sustained	Rapid and intense

Olympic marathon runners have a high proportion of slow twitch fibres in their muscles.

Explain how the features of slow twitch fibres, shown in the table, are of benefit to an Olympic marathon runner.

(4)

(c) The graph shows the rate of heat loss at rest and during exercise.



Comment on the rate of heat loss during exercise.

(2)

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(Total for Question 1 = 8 marks)

